
When Does Retrieval-Augmented Generation Help?

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Abstract

Retrieval-Augmented Generation (RAG) augments language models with non-parametric external memory, but its benefits depend on many pipeline decisions that are rarely studied in controlled isolation. We present a twelve-experiment ablation on TriviaQA ($n = 750$ questions, pooled Wikipedia/web corpus) using Flan-T5-base as the generator, covering retrieval method, chunk granularity, retrieval depth, prompt template, oracle and cross-encoder re-ranking upper bounds, external distractor robustness, controlled distraction at fixed recall, forced retrieval miss, and generator scale. Our main findings are: (i) RAG provides a massive benefit over the parametric baseline ($\Delta\text{EM} = +0.551$, Holm $p < 10^{-4}$) because Flan-T5-base genuinely lacks long-tail factual knowledge; (ii) the generator, not the retriever, is the binding constraint at this scale—retrieval method, embedding model, and cross-encoder re-ranking are all statistically indistinguishable under Holm correction; (iii) with the answer guaranteed present, each additional hard topical distractor monotonically reduces EM ($0.673 \rightarrow 0.558$ across $N = 0\text{--}8$ distractors); and (iv) wrong-but-topical retrieved context primes parametric recall ($+0.102$ EM over an empty-context baseline), so a retrieval miss is costly relative to correct retrieval but does not drag the model below its parametric floor.

1 Introduction

Large language models (LLMs) encode substantial world knowledge during pre-training, but this *parametric memory* has well-known failure modes: facts can be outdated, hallucinated, or simply absent for long-tail entities [1]. Retrieval-Augmented Generation (RAG) addresses this by augmenting the generator with a non-parametric corpus—retrieving relevant passages at inference time and conditioning the model on them.

Despite RAG’s strong empirical track record, its benefits are not uniform and the contribution of individual components remains poorly understood. Retrieval can be noisy or fail entirely; irrelevant passages can *distort* the generator; and chunking strategy, embedding model, and retrieval depth are typically tuned by intuition rather than evidence. Prior work establishes that RAG helps on knowledge-intensive tasks [1], that how passages are integrated matters [2], and surveys the full design space [3]—but causal component-level ablations under a consistent evaluation protocol on the same benchmark are rare.

We address this gap with a controlled twelve-experiment study. Our key contributions are:

- A comprehensive ablation spanning retrieval family (BM25 / TF-IDF / Dense), embedding model, chunk granularity, retrieval depth k , prompt template, oracle upper bound, cross-encoder re-ranking, external distractor robustness, and generator scale.

- Statistical evidence (Holm-corrected paired tests) that the *generator* is the binding constraint: retrieval method, embedding model, and re-ranking are all indistinguishable within ± 0.02 EM—a *positive* “they tie” conclusion, not a power-limited one.
- The first clean causal estimate of when RAG hurts: pinning recall at 1 and adding hard topical distractors causes a monotone 11.5 pp EM drop ($0.673 \rightarrow 0.558$), entirely attributable to generator distraction.
- Evidence that topical retrieval misses prime parametric recall ($+0.102$ EM over an empty-context baseline, Holm $p < 10^{-4}$), and that the RAG–No-RAG gap begins to shrink as model scale grows.

2 Method

System. We implement a standard retrieve-then-read pipeline. A retriever selects the top- k passages from a corpus; these are concatenated into a prompt (question repeated before and after the context for robustness to truncation); and the generator produces a short answer. We evaluate three retrieval families: **BM25** (Okapi BM25 on a shared NFKD-normalised token stream), **TF-IDF** (TfidfVectorizer, sublinear TF, same tokeniser as BM25), and **Dense** (sentence-transformers all-MiniLM-L6-v2, 384-dim, cosine similarity). For the embedding ablation (Exp. 10) and cross-encoder re-ranking (Exp. 7) we additionally use bge-small-en-v1.5, e5-small-v2, and ms-marco-MiniLM-L-6-v2. The generator is **Flan-T5-base** [5] (≈ 250 M parameters), with greedy decoding, a 1024-token input budget, and middle-truncation to preserve the answer cue.

Dataset and corpus. We use **TriviaQA rc** [4], sampling 750 questions from the validation split (seed 42). TriviaQA was chosen because the primary references for this work—Lewis et al. [1] and Izacard and Grave [2]—both evaluate on TriviaQA, enabling direct comparison with those baselines. TriviaQA provides Wikipedia entity pages and web search hits for each question; pooling all documents across questions gives a realistic retrieval setting where the gold evidence is always present but must be located among $\sim 2,500$ documents (including 2,000 topic-agnostic Simple English Wikipedia distractors added to widen the noise floor). Documents are chunked with a fixed word-window (default 128 words, 32-word overlap). Gold answers are SQuAD-normalised (lowercase, strip articles/punctuation) and a prediction is correct if it matches any gold alias.

Evaluation. We report **Exact Match (EM)**, **Token F1**, and **Retrieval Recall@ k** (a whole-word token-span match, avoiding spurious partial-word substring hits). All metrics carry 95% percentile bootstrap CIs (1,000 resamples). For pairwise comparisons we use **McNemar’s exact test** on EM and a **paired bootstrap** on score deltas, with **Holm–Bonferroni correction** across the family of nine contrasts. Paired tests cancel question-to-question difficulty variance, providing substantially more power than comparing marginal CIs.

3 Experiments and Results

RAG vs. No-RAG (Exp. 5). RAG provides a massive benefit over the parametric floor (Table 1, top): $\Delta\text{EM} = +0.551$, Holm $p < 10^{-4}$ (discordant 419/6 questions). The No-RAG EM of 0.049 confirms that Flan-T5-base genuinely lacks these long-tail facts—the model produces confidently-shaped but incorrect entity strings (e.g., `henry viii` for “PM after Balfour”). Of 750 questions, **411 benefit**, **319 tie**, and only **6 are hurt** by retrieval; the 6 “hurts” are questions the model already answered correctly from memory and where retrieved context distracted it.

Retrieval pipeline: method, chunk size, and depth (Exps. 1–4). All three retrieval families are statistically indistinguishable (Table 1, middle; Holm $p = 1.0$ for all three pairs): Dense retrieval achieves the highest Recall@5 (0.865) but not the highest EM, confirming that residual error is generator-side. At a fixed 1024-token budget (Exp. 2), smaller chunks win: 64-word chunks (≈ 8 packed per prompt) achieve EM 0.596 vs. 0.517 for 512-word chunks (≈ 1.2 per prompt), since large chunks risk losing the entire context window on a single retrieval miss. Retrieval depth k (Exp. 3, chunk = 48 words) shows monotone improvement through $k = 10$ (EM: $0.433 \rightarrow 0.583$) with no distraction penalty—consistent with distraction only materialising once the answer is already

Table 1: Core results. All experiments use Flan-T5-base, chunk=128 words, $k=5$, instructed prompt, $n=750$ (Exp. 10: $n=400$). “Holm p ” is the McNemar p -value adjusted across the nine-contrast family; “(ref)” marks the comparison baseline.

Condition	EM	F1	R@5	Holm p
<i>RAG vs. No-RAG (Exp. 5)</i>				
No-RAG (parametric only)	0.049	0.100	—	
RAG (Dense, $k=5$)	0.600	0.668	0.865	$<10^{-4}$
<i>Retriever comparison (Exp. 1)</i>				
BM25	0.597	0.670	0.859	1.00
TF-IDF	0.603	0.673	0.852	1.00
Dense (MiniLM)	0.600	0.668	0.865	(ref)
<i>Embedding model ablation (Exp. 10)</i>				
Dense (MiniLM)	0.623	0.680	0.877	(ref)
Dense (bge-small)	0.620	0.673	0.905	—
Dense (e5-small)	0.610	0.663	0.915	—
<i>Oracle upper bound (Exp. 6)</i>				
RAG (Dense, $k=5$)	0.600	0.668	0.865	(ref)
Oracle (gold forced in)	0.619	0.684	0.999	0.0007

guaranteed present. Prompt template (Exp. 4) has no significant effect: concise vs. instructed differ by only 0.016 EM (Holm $p=0.18$).

Retrieval quality ceiling: oracle, re-ranking, and embedding model (Exps. 6, 7, 10). Three experiments probe whether better retrieval quality can raise EM, and all point to the same conclusion: the generator is the binding constraint. The oracle upper bound (Exp. 6, Table 1, bottom) forces the gold chunk in for the 13% of questions that Dense misses, yielding only +0.019 EM (Holm $p=0.0007$)—small but reliable. Cross-encoder re-ranking (Exp. 7, ms-marco-MiniLM-L-6-v2, top-50→top-5) raises Recall@5 by 0.017 but the EM gain (+0.017, Holm $p=0.73$) is not significant. The embedding model ablation (Exp. 10, $n=400$) reveals the same pattern: e5-small reaches Recall@5=0.915 (highest) yet EM=0.610 (lowest), while MiniLM has Recall@5=0.877 but EM=0.623 (highest)—a stronger retriever finds more gold chunks; the generator extracts no better answers from them.

Distractor robustness and retrieval failure (Exps. 8, 9, 11). Experiment 8 sweeps the external distractor count ($\{0, 500, 2000, 5000\}$ Simple-Wikipedia articles): Recall@5 and EM are exactly flat across a $2.8\times$ corpus-size range, confirming that topic-agnostic noise never out-ranks gold pages. To isolate generator distraction from retrieval failure, Experiment 9 pins the gold chunk at rank 1 (Recall@ $k=1$ throughout) and adds $N \in \{0, 1, 2, 4, 8\}$ hard topical non-gold chunks. EM drops **monotonically** from 0.673 ($N=0$, single-passage ceiling) to 0.558 ($N=8$)—a fall of 11.5 pp attributable entirely to distraction (Figure 1, left). Experiment 11 (forced retrieval miss, Recall@ $k=0$ by construction) closes the analysis from the other side: wrong-but-topical context scores EM 0.159—far below RAG (0.600) but well above no-context baselines (0.049–0.057). Decomposing: the instructed-prompt scaffold alone adds +0.008 EM over the bare parametric baseline; topical-but-wrong context adds a further +0.102 EM (Holm $p < 10^{-4}$). A retrieval miss therefore primes parametric recall rather than dragging the model below its floor.

Generator scale (Exp. 12). Figure 1 (right) sweeps Flan-T5 small/base/large (80M / 250M / 780M) on 150 questions. The No-RAG parametric floor rises monotonically with scale (0.027→0.053→0.113), while RAG rises more slowly (0.467→0.600→0.633) toward the retrieval/generation ceiling. The RAG–No-RAG gap **peaks at base** ($\Delta = 0.547$) and **begins to shrink at large** ($\Delta = 0.520$), consistent with retrieval helping most when parametric memory is weakest. The large model is also the only one where RAG occasionally hurts (3/150 questions) as growing parametric knowledge conflicts with retrieved context.

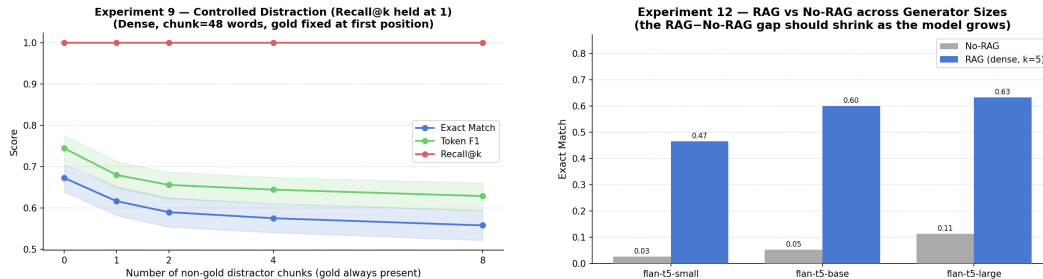


Figure 1: **Left (Exp. 9):** EM and F1 decline monotonically as hard topical distractors are added alongside a guaranteed gold chunk (Recall@k = 1 throughout). The $N = 0$ point (gold chunk only) is the single-passage generator ceiling. **Right (Exp. 12):** RAG vs. No-RAG EM across Flan-T5 scales. The No-RAG parametric floor rises monotonically; the RAG–No-RAG gap peaks at base and begins to shrink at large, confirming that retrieval helps most when parametric memory is weakest.

4 Conclusion

We conducted a twelve-experiment controlled ablation of a RAG pipeline on TriviaQA. Four contrasts survive Holm correction: RAG vs. No-RAG (+0.551 EM, the dominant effect); both comparisons of forced-miss against no-context baselines (+0.102–0.110 EM, topical priming); and Oracle vs. RAG (+0.019 EM, a small but reliable retrieval-failure component first resolved at $n = 750$). Retrieval method, embedding model, and re-ranking are all Holm-non-significant within ± 0.02 EM with large discordant counts—a positive “they tie” result. Experiment 9 provides the clearest causal picture of when RAG hurts: each additional topical distractor reduces EM monotonically even when the answer is guaranteed present. Taken together, the results suggest that for small instruction-tuned generators, RAG effort is better invested in the generator than in retrieval quality.

Limitations. The study uses small encoder-decoders (up to 780M parameters) on a corpus much smaller and less adversarial than open-domain Wikipedia. A DPR model fine-tuned on TriviaQA would likely widen the sparse–dense gap currently observed as a tie. Much larger generators (LLaMA-class) would raise the parametric floor further and shrink the RAG benefit; multi-hop questions and adversarially constructed distractors are left to future work.

References

- [1] Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., ... Kiela, D. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *NeurIPS 2020*.
- [2] Izacard, G., & Grave, E. (2021). Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering. *EACL 2021*.
- [3] Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., ... Wang, H. (2024). Retrieval-Augmented Generation for Large Language Models: A Survey. *arXiv:2312.10997*.
- [4] Joshi, M., Choi, E., Weld, D., & Zettlemoyer, L. (2017). TriviaQA: A Large Scale Distantly Supervised Challenge Dataset for Reading Comprehension. *ACL 2017*.
- [5] Chung, H. W., Hou, L., Longpre, S., Zoph, B., Tay, Y., Fedus, W., ... Wei, J. (2022). Scaling Instruction-Finetuned Language Models. *arXiv:2210.11416*.
- [6] Robertson, S., & Zaragoza, H. (2009). The Probabilistic Relevance Framework: BM25 and Beyond. *Foundations and Trends in Information Retrieval*, 3(4).

A Additional Figures

Figures 2–16 show all fifteen experiment plots from the report-scale run ($n = 750$, $n = 400$ for Exp. 10, $n = 150$ for Exp. 12) on GPU (HPC). Error bars are 95% percentile bootstrap CIs (1,000 resamples).

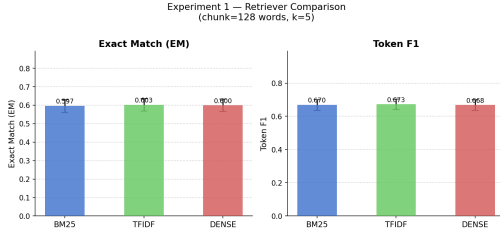


Figure 2: Exp. 1 — Retriever comparison (BM25 / TF-IDF / Dense). All three are statistically indistinguishable (Holm $p=1.0$ for all pairs).

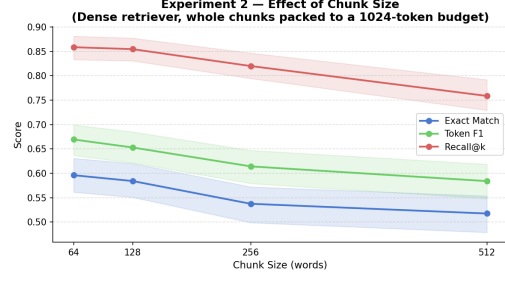


Figure 3: Exp. 2 — Chunk-size ablation at a fixed 1024-token budget. Smaller chunks pack more passages and achieve higher recall and EM.

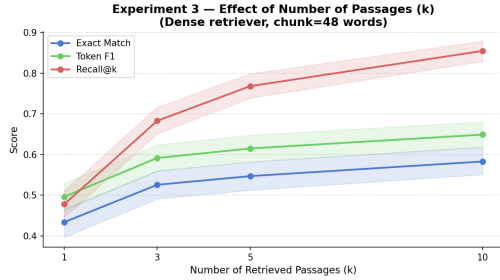


Figure 4: Exp. 3 — Retrieval depth sweep ($k \in \{1, 3, 5, 10\}$, chunk = 48 words). EM and Recall@ k both increase monotonically; no distraction penalty appears at standard retrieval.

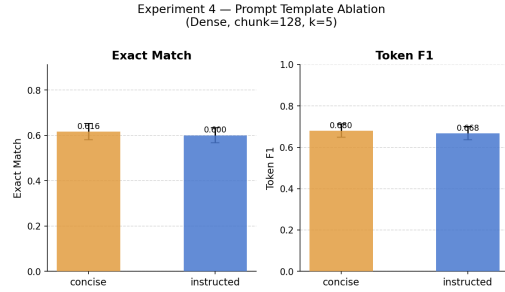


Figure 5: Exp. 4 — Prompt template: concise vs. instructed. Difference is 0.016 EM (Holm $p=0.18$, non-significant).

B Complete Results Tables

Table 2: Exp. 2 — Chunk size at fixed 1024-token budget (Dense, $n = 750$, 30 candidates packed greedily).

Chunk (words)	Mean packed k	EM	Token F1	Recall@ k
64	8.27	0.596	0.669	0.859
128	4.29	0.584	0.653	0.855
256	2.14	0.537	0.614	0.820
512	1.19	0.517	0.584	0.759

C Statistical Significance Tests

Table 8 reports all nine pairwise contrasts with raw and Holm-Bonferroni-adjusted p -values. Four contrasts survive correction (bold): RAG vs. No-RAG, both Forced-Miss contrasts, and Oracle vs. RAG.

D Implementation Details

Chunking. Fixed word-window chunking with 32-word overlap. For a document of W words and chunk size c : chunk i spans words $[i(c - 32), i(c - 32) + c]$.

Generation. Flan-T5-base with greedy decoding (`do_sample=False`, `num_beams=1`, `max_new_tokens=32`). Long prompts are *middle-truncated*: prefix (question) and suffix (answer cue) are preserved; only the context body is trimmed. All generation is cached to `data/cache/generation_cache.json` keyed by MD5 of `model_name + rendered_prompt`, so different generator sizes (Exp. 12) never collide.

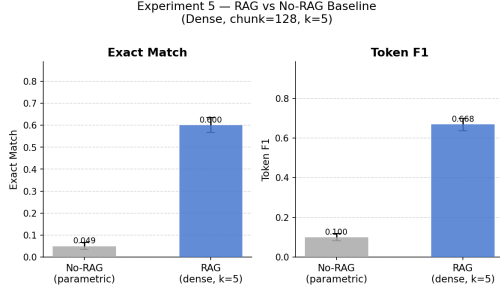


Figure 6: Exp. 5 — RAG vs. No-RAG. $\Delta\text{EM} = +0.551$ (Holm $p < 10^{-4}$; 411 helps / 6 hurts / 319 ties out of 750 questions).

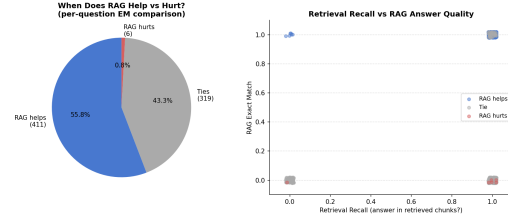


Figure 7: Exp. 5 error analysis: per-question RAG helps / hurts / ties breakdown and scatter of Recall@5 vs. ΔEM .

Experiment 5 — Qualitative Examples: When RAG Helps vs Hurts

category	Question (truncated)	Real	No-RAG pred	RAG pred
RAG helps	What general name is given to a rotating...	rotation	4)	rotation
RAG helps	Which object and feature operates in a...	radioactive	radioactive	radioactive
RAG helps	What is the primary cause of depression?	depression	depression	depression
RAG helps	How many "Rings" are there in the leg...	four	four	four
RAG hurts	Scientist Ernest Rutherford once said ab...	physics	physics	physics or atomic science
RAG hurts	A member of which profession would not...	clergymen	clergymen	clergymen

Figure 8: Exp. 5 qualitative examples where RAG helps and where it hurts.

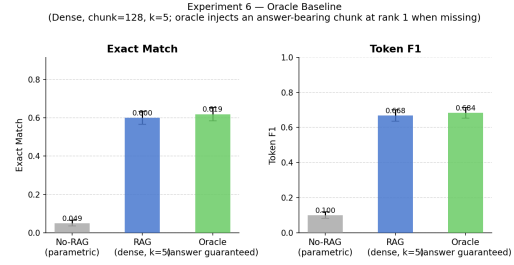


Figure 9: Exp. 6 — Oracle vs. RAG vs. No-RAG. Oracle gain: $+0.019$ EM (Holm $p = 0.0007$), quantifying the small reliable retrieval-failure component.

Dense retrieval. Query and passage embeddings are L2-normalised; retrieval is by dot product (equivalent to cosine similarity). Passage embeddings are cached per chunk-size to avoid re-encoding across experiments. For Exp. 10, each model receives its recommended asymmetric query/passage instruction prefixes (e5: "query: "; "passage: "; bge: search instruction on the query side; MiniLM: no prefix).

Reproducibility. All random processes use fixed seeds (corpus sampling seed 42, bootstrap seed 42). Running the notebook twice produces identical results. Expected end-to-end runtime at $n = 750$: approximately 4–8 h on CPU (significantly less on GPU/MPS with embedding caching).

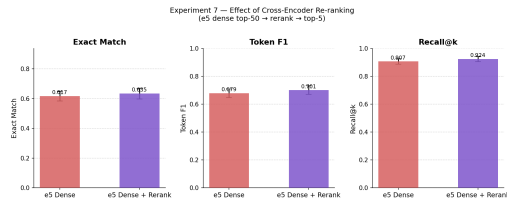


Figure 10: Exp. 7 — Cross-encoder re-ranking (e5 first stage, top-50→top-5). Recall@5 rises by 0.017 but EM gain (+0.017) is not significant (Holm $p=0.73$).

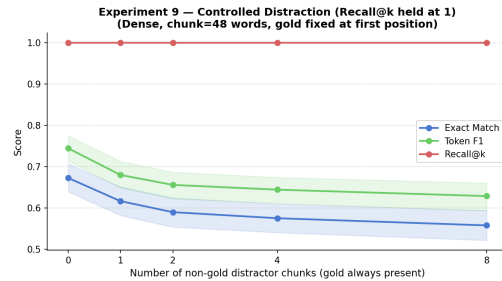


Figure 12: Exp. 9 — Controlled distraction (Recall@ $k=1$ throughout). Adding hard topical distractors drives EM monotonically from 0.673 ($N=0$) to 0.558 ($N=8$).

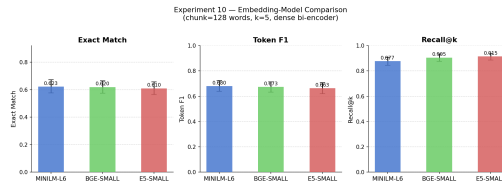


Figure 14: Exp. 10 — Embedding model ablation ($n = 400$). e5-small achieves the highest Recall@5 (0.915) yet the lowest EM (0.610); MiniLM has the lowest recall but the highest EM (0.623).

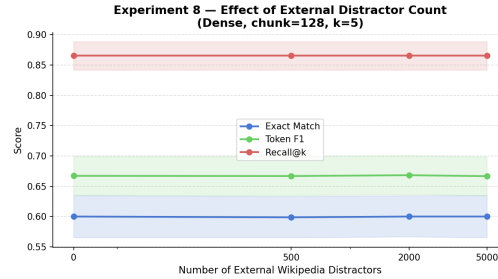


Figure 11: Exp. 8 — External distractor count sweep ($\{0, 500, 2000, 5000\}$ Simple-Wikipedia articles). Recall@5 and EM are exactly flat across a $2.8\times$ corpus-size range: topic-agnostic distractors never out-rank gold pages.

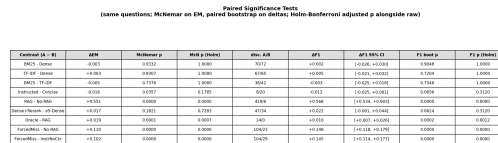


Figure 13: Paired significance table (McNemar on EM, paired bootstrap on deltas; raw and Holm-adjusted p -values for all nine contrasts).

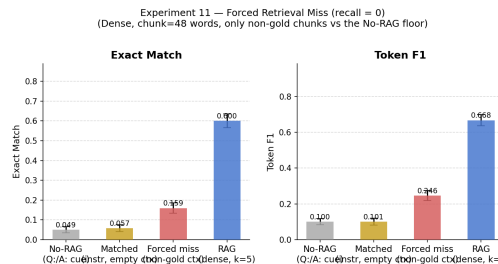


Figure 15: Exp. 11 — Forced retrieval miss (Recall@ $k = 0$). Topical-but-wrong context adds +0.102 EM over the empty-context baseline (Holm $p < 10^{-4}$) via parametric priming.

Table 3: Exp. 3 — Retrieval depth k (Dense, chunk = 48 words, $n = 750$; full range fits 1024-token budget untruncated).

k	EM	Token F1	Recall@ k
1	0.433	0.496	0.477
3	0.525	0.591	0.683
5	0.547	0.615	0.768
10	0.583	0.649	0.855

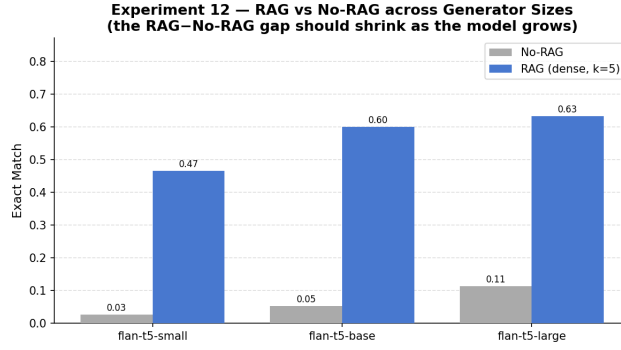


Figure 16: Exp. 12 — Generator-scale sweep ($n = 150$). No-RAG floor rises with scale ($0.027 \rightarrow 0.053 \rightarrow 0.113$); RAG–No-RAG gap peaks at base ($\Delta = 0.547$) and shrinks at large ($\Delta = 0.520$).

Table 4: Exp. 9 — Controlled distraction (Recall@ $k = 1$ by construction; gold chunk pinned at rank 1, $n \approx 750$, chunk = 48 words).

N distractors	EM	Token F1	Recall@ k
0	0.673	0.745	1.00
1	0.617	0.680	1.00
2	0.590	0.656	1.00
4	0.575	0.645	1.00
8	0.558	0.629	1.00

Table 5: Exp. 11 — Forced retrieval miss ($n \approx 750$, chunk = 48 words; Recall@ $k = 0$ by construction for the forced-miss arm).

Condition	EM	Token F1	Recall@ k
No-RAG (bare Q: /A: cue)	0.049	0.100	—
Matched (instructed prompt, <i>empty</i> context)	0.057	0.101	—
Forced miss (instructed prompt, non-gold chunks)	0.159	0.246	0.00
RAG (Dense, $k = 5$, Exp. 5)	0.600	0.668	0.865

Table 6: Exp. 12 — Generator-scale sweep ($n = 150$; trend demonstration on a representative subset).

Generator	No-RAG EM	RAG EM	Δ EM	Helps / Hurts / Ties
flan-t5-small (≈ 80 M)	0.027	0.467	+0.440	67 / 1 / 82
flan-t5-base (≈ 250 M)	0.053	0.600	+0.547	83 / 1 / 66
flan-t5-large (≈ 780 M)	0.113	0.633	+0.520	81 / 3 / 66

Table 7: Exp. 8 — External distractor count sweep (Dense, chunk = 128, $k = 5$, $n = 750$; corpus sizes are approximate).

Distractors	Corpus docs	EM	Token F1	Recall@5
0	$\sim 2,800$	0.600	0.667	0.865
500	$\sim 3,300$	0.599	0.667	0.865
2000	$\sim 4,800$	0.600	0.668	0.865
5000	$\sim 7,800$	0.600	0.667	0.865

Table 8: Paired significance results ($n=750$). McNemar’s exact test on EM; paired bootstrap on F1 deltas (10,000 resamples). Holm–Bonferroni correction across all nine contrasts. Discordant counts are (A better / B better).

Contrast (A – B)	ΔEM	Disc.	McN raw	McN Holm	ΔF1	95% CI	F1 Holm
BM25 – Dense	−0.003	70/72	0.933	1.000	+0.002	[−0.026, +0.030]	1.000
TF-IDF – Dense	+0.003	67/65	0.931	1.000	+0.005	[−0.021, +0.032]	1.000
BM25 – TF-IDF	−0.005	38/42	0.738	1.000	−0.003	[−0.025, +0.018]	1.000
Instructed – Concise	−0.016	8/20	0.036	0.179	−0.012	[−0.025, +0.001]	0.312
RAG – No-RAG	+0.551	419/6	< 0.001	< 0.001	+0.568	[+0.534, +0.603]	< 0.001
Dense+Rerank – e5-Dense	+0.017	47/34	0.182	0.728	+0.022	[−0.001, +0.044]	0.312
Oracle – RAG	+0.019	14/0	0.0001	0.0007	+0.016	[+0.007, +0.026]	0.001
ForcedMiss – No-RAG	+0.110	104/23	< 0.001	< 0.001	+0.148	[+0.118, +0.179]	< 0.001
ForcedMiss – InstrNoCtx	+0.102	104/29	< 0.001	< 0.001	+0.145	[+0.114, +0.177]	< 0.001